Separation and Purification Technology 411 (2026) 139525

Contents lists available at ScienceDirect

Separation and Purification Technology

journal homepage: www.elsevier.com/locate/seppur

Green extraction and stabilization of bioactive compounds from annatto (*Bixa orellana* L.) seeds using hydrophobic natural deep eutectic solvents

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ARTICLE INFO

Editor name: M Freire

Keywords:
Extraction of carotenoids
Bixin
Geranylgeraniol
Antioxidant activity
Antimicrobial activity
Chemical stability

ABSTRACT

Hydrophobic natural deep eutectic solvents (HNADES) have emerged as promising green alternatives to conventional organic solvents for the extraction of lipophilic bioactive compounds. In this study, saturated fatty acid–based HNADES were employed as multifunctional media for the simultaneous extraction and stabilization of bioactive compounds from annatto (*Bixa orellana* L.) seeds, with particular emphasis on bixin and geranylgeraniol. HNADES composed of dodecanoic acid combined with decanoic, octanoic, or heptanoic acids were formulated and physicochemically characterized. Extraction parameters were optimized using a one-factor-at-a-time approach by evaluating solvent composition, seed concentration, temperature, and extraction time. Among the evaluated systems, C12:C8 (1:3 M ratio) showed the highest extraction efficiency, yielding greater total carotenoid content and antioxidant capacity than conventional solvents (ethanol and methanol). High-performance liquid chromatography confirmed significantly higher concentrations of bixin and geranylgeraniol in the optimized extract. Furthermore, the HNADES-based system improved the stability of carotenoids, antioxidant activity, and color over 90 days of storage, exhibiting substantially lower degradation rates than the control extracts, likely due to the physicochemical properties of the eutectic matrix. The optimized extract also demonstrated relevant antibacterial activity against selected foodborne bacteria. Together, these findings demonstrate that fatty acid-based HNADES act as multifunctional and sustainable media that simultaneously enhance extraction efficiency, improve the stability of oxidation-sensitive lipophilic bioactive compounds, and support the valorization of annatto seeds through a green extraction strategy.

1. Introduction

Bixa orellana L., commonly known as annatto in the United States and urucum in Brazil, is a medicinal tree native to tropical America, belonging to the Bixaceae family, that has attracted considerable industrial and scientific interest due to the high concentration of bioactive compounds in its seeds [1]. Annatto seeds are rich in carotenoids, terpenoids [2], and phenolic compounds [3], primarily used as natural colorants in food, pharmaceuticals, and cosmetics, mainly because bixin constitutes over 80% of the carotenoids in the seed coat [1,4]. Bixin, a long-chain orange-red pigment, is prone to oxidative degradation and isomerization during extraction [4,5]. Besides its coloring properties, it possesses antioxidant, antimicrobial, anti-inflammatory, and anticancer effects [6–8]. Annatto seeds also contain terpenoids like geranylgeraniol, an acyclic diterpene alcohol essential for carotenoid biosynthesis ([9]; Martinez [10]). Geranylgeraniol exhibits similar antioxidant, antimicrobial, anti-inflammatory, and anticancer properties [9], while tocopherols, tocotrienols [7], and phenolic compounds are present in smaller amounts [3].

Despite the recognized health benefits of these compounds, their practical application remains limited because extraction still largely relies on organic solvents, given their pronounced lipophilicity. Common solvents include hexane [8], acetone [11], ethyl acetate, ethanol, and methanol [12], as well as alkaline treatments at the industrial scale [13]. However, these approaches generate hazardous waste, present volatility, and flammability risks, and may restrict the broader use of annatto extracts for therapeutic purposes [14]. Consequently, the development of green extraction technologies is essential to expand the

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https://doi.org/10.1016/j.seppur.2026.139525
Received 15 April 2026; Received in revised form 3 July 2026; Accepted 26 July 2026
Available online 27 July 2026

applicability of these bioactive compounds.

Within the Green Chemistry framework, natural deep eutectic solvents (NADES) have emerged as promising alternatives, predominantly as hydrophilic systems, while hydrophobic NADES (HNADES) have been more recently developed and explored for extracting bioactive compounds from plant matrices [15,16]. These multicomponent systems have lower melting points than their individual components due to intermolecular interactions, remaining liquid at 25 °C for easier handling [15,17–19]. The fatty acid-based HNADES employed in the present study have also been previously thermodynamically characterized through solid-liquid equilibrium (SLE) phase diagrams, confirming their eutectic behavior and supporting their classification as deep eutectic solvents [20]. Additionally, NADES are typically non-toxic, safe, and cost-effective, often made from food-grade materials [21,22].

In HNADES, long-chain hydrophobic components, particularly saturated fatty acids, are typically employed to enable the efficient extraction of lipophilic compounds [16,19]. Fatty acid-based HNADES composed of medium- and long-chain fatty acids have been successfully applied to the extraction of lipophilic bioactive compounds from different biomass sources, such as β-carotene from pumpkin and orange peel [8,16,23], lycopene and other carotenoids from tomato [24,25], lutein, zeaxanthin and other carotenoids from microalgae biomass [26], carotenoids [26], and lycopene, lutein, zeaxanthin, γ-carotenoid, and other carotenoids from canistel (*Lucuma nervosa*) pulp [27], among others. These studies highlight the strong affinity of fatty acid-based HNADES for lipophilic compounds and support their potential as green alternatives to conventional organic solvents. In addition to their extraction capacity, these solvents may enhance the stability of bioactive compounds through physical interactions and their relatively high viscosity, which can reduce oxygen diffusion and limit oxidative degradation [8,28].

To the best of our knowledge, no studies have investigated the use of saturated fatty acid-based HNADES for the extraction of bioactive compounds from annatto seeds, particularly bixin and geranylgeraniol. Beyond introducing these HNADES as extraction media for annatto seeds, this study provides an integrated evaluation of their technological potential by optimizing the extraction conditions, quantifying the major bioactive compounds (bixin and geranylgeraniol), assessing the stability of the extracts during storage, evaluating their antimicrobial activity, and examining the greenness of the proposed extraction process using the AGREEprep metric. Therefore, this work contributes not only by expanding the application of HNADES to a new biomass, but also by demonstrating their multifunctional role as extraction, stabilization, and sustainable processing media for lipophilic bioactive compounds. Therefore, this study aimed to optimize the extraction conditions of annatto seed bioactives using HNADES composed of dodecanoic, decanoic, octanoic, and heptanoic acids based on total carotenoid content and antioxidant capacity, quantify bixin and geranylgeraniol in the optimized extracts, and evaluate the stability of carotenoids, antioxidant activity, and color over 90 days of storage using methanol- and ethanol-based extracts as controls. The antimicrobial activity of the optimized extract was also assessed.

2. Material and methods

2.1. Material

The annatto seeds were purchased at a local market in São Carlos-SP, Brazil. Dodecanoic acid (Lauric acid, $C_{12}H_{24}O_2$, 200 g mol^{-1}, analytical grade), decanoic acid (Capric acid, $C_{10}H_{20}O_2$, 172.26 g mol^{-1}, ≥ 98% purity), octanoic acid (Caprylic acid, $C_8H_{16}O_2$, 144.21 g mol^{-1}, analytical grade), and heptanoic acid ($C_7H_{14}O_2$, 130.18 g mol^{-1}, ≥ 99% purity), were obtained from Êxodo Científica (Sumaré-SP, Brazil), Neon (Suzano-SP, Brazil), ACS Científica (Sumaré-SP, Brazil), and Sigma-Aldrich (St. Louis-MO, USA), respectively. Acetone P.A., ethanol P.A., hexane P.A. and methanol P.A. were obtained from Êxodo Científica (Sumaré-SP, Brazil). Butylhydroxytoluene and tetrahydrofuran P.A. were obtained from Neon (Suzano-SP, Brazil). Analytical reagents, such as gallic acid (97.5–102.5% purity), bixin (≥ 90% purity), geranylgeraniol (≥ 85% purity), β-carotene (≥ 95% purity), Nile red dye (≥ 97% purity), (±)-6-hydroxy-2,5,7,8-tetramethylchromane-2-carboxylic acid (Trolox, ≥ 97% purity), and 2,2′-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid (ABTS, ≥ 98% purity) were purchased from Sigma-Aldrich (St. Louis-MO, USA).

2.2. Formulation of HNADES

HNADES were formulated by mixing dodecanoic acid with decanoic acid, octanoic acid or heptanoic acid in molar ratios of 1:2, 1:3 and 1:2, respectively. The HNADES components were homogenized in a 240 W ultrasonic bath (LT-210, LimaTec, São Carlos-SP, Brazil) at 60 °C for 20 min until a homogeneous mixture was obtained.

2.3. Characterization of HNADES

2.3.1. Density and apparent acidity

The density of HNADES and conventional solvents was determined using a pycnometer, while the apparent acidity of HNADES was estimated using pH indicator strips due to their hydrophobic nature and low electrical conductivity, which can hinder reliable potentiometric measurements with conventional glass electrodes. Therefore, the reported values should be interpreted as approximate indicators of acidity rather than absolute measurements. The pH of conventional solvents was measured using a portable pH meter (K39–220, Kasvi, Pinhais-SP, Brazil) at 25 °C.

2.3.2. Viscosity

The apparent viscosity of HNADES was obtained using a controlled stress rheometer (AR-1000 N, TA Instruments, New Castle-DE, USA) with a Peltier system for temperature control, using a cone-plate geometry (cone angle of 2°, 40 mm diameter, and gap of 55 μm). Apparent viscosity was determined over a temperature range of 20–75 °C at a constant shear rate of 1 s^{-1} using a controlled-stress rheometer [21].

2.3.3. Polarity

The polarity of HNADES and conventional solvents was estimated by the solvatochromic method [29], which evaluates the change in the transition energy (E_{NR}) of the Nile red dye solvated in different solvents. HNADES and conventional solvents were mixed with a Nile red dye ethanolic solution 1 mg mL^{-1} (1:200 dye:solvent) and wavelength scanning was performed from 800 to 200 nm. The wavelength of the maximum absorbance value (λ_{max}) was used to calculate the E_{NR} (Eq. (1)), which is an indirect polarity parameter.

$$E_{NR}\left(kcal\,mol^{-1}\right) = \frac{28591.44}{\lambda_{max}} \quad (1)$$

2.3.4. Differential scanning calorimetry (DSC)

The thermal properties of HNADES and their respective forming fatty acids were determined using a Differential scanning calorimeter – DSC (Discovery DSC 250 Auto, TA Instruments, New Castle-DE, USA) with control and analysis software package TRIOS software v5.8.1.14 (TA Instruments, New Castle-DE, USA), and equipped with a Discovery Refrigerated Cooling System – RCS90 accessory. Samples were placed in hermetically sealed aluminum pan and heated from −70 °C to 70 °C at 5 °C min^{-1} in an inert atmosphere (45 mL min^{-1} N_2) [19]. Melting temperatures and enthalpies were calculated directly from the thermal curves using the equipment software.

2.4. Optimization of HNADES-annatto seed extract production

To optimize the extraction conditions for bioactive compounds from annatto seeds using HNADES as solvents, a one-factor-at-a-time (OFAT) screening was conducted. The evaluated factors included HNADES composition (C12:C10 1:2, C12:C8 1:3, and C12:C7 1:2), solid-to-solvent ratio (100, 150, 200, and 300 mg annatto seeds mL^{-1} solvent), extraction temperature (40, 50, 60, and 70 °C), and extraction time (40, 50, 60, and 70 min) (Fig. 1). For this, the extraction conditions optimized by Jayakumar et al. [30] to extract carotenoids from annatto seeds using conventional organic solvents were adopted as initial fixed conditions: 70 °C, 30 min and 150 mg annatto seed mL^{-1} solvent. The HNADES-annatto seed extracts were produced in 240 W ultrasound bath (LT-210, LimaTec, São Carlos-SP, Brazil). Each optimized condition was chosen based on the results of total carotenoid content, and antioxidant capacity by the ABTS free radical scavenging assay (ABTS assay), described in the following sections.

2.5. Characterization of HNADES-annatto seed extracts

2.5.1. Total carotenoid content

Total carotenoid content of HNADES-annatto seed extract was determined according to methodology described by Lüdtke et al. [31]. First, 2.5 mL of HNADES-annatto seed extracts, properly diluted in ethanol, were vortexed for 30 s with 5 mL of 0.01% butylated hydroxytoluene in acetone. Next, 5 mL of hexane was added and vortexed for 10 s. To complete the final volume of 25 mL, 12.5 mL of distilled water was added, and the mixture was vortexed for another 10 s. The samples were then centrifuged at 5000 rpm and 4 °C for 5 min in a high-speed refrigerated centrifuge (CR22GIII, Hitachi, Tokyo, Japan). The supernatant hexane phase was collected and analyzed using a UV-VIS spectrophotometer at 450 nm. Total carotenoid content was expressed as µg of β-carotene g^{-1} of annatto seed, based on a β-carotene standard curve (0.06 to 0.46 µg mL^{-1}) with a determination coefficient of 0.9945.

2.5.2. Antioxidant capacity

The antioxidant capacity of HNADES-annatto seed extracts was determined by ABTS free radical scavenging ($ABTS^{\bullet+}$) assay [32]. The $ABTS^{\bullet+}$ assay was selected because it is a mixed-mode method capable of evaluating antioxidant activity through both single electron transfer (SET) and hydrogen atom transfer (HAT) mechanisms, which are involved in the antioxidant action of carotenoids. Furthermore, $ABTS^{\bullet+}$ assay is suitable for the analysis of both hydrophilic and lipophilic compounds, making it particularly appropriate for carotenoid-rich extracts [33].

ABTS free radical ($ABTS^{\bullet+}$) was obtained after a 16 h reaction in the dark between 440 µL of 140 mM potassium persulfate aqueous solution and 7 mM ABTS aqueous solution completed to a final volume of 25 mL. Subsequently, the $ABTS^{\bullet+}$ solution formed was diluted approximately 100 times in ethanol, to obtain an absorbance of 0.70 ± 0.05 at 734 nm (UV–Vis spectrophotometer, UV-M51, BEL® Engineering, Monza-MB, Italy). The test was then performed by mixing 30 µL of HNADES-annatto seed extract properly diluted in ethanol with 3 mL of the $ABTS^{\bullet+}$ solution diluted in ethanol, and the absorbance was taken at 734 nm after 20 min of reaction. The sample blank was prepared by mixing 30 µL of diluted sample and 3 mL of ethanol. The antioxidant capacity was expressed as mg Trolox equivalent (TE) g^{-1} annatto seed, using a Trolox standard curve (0.025 to 0.470 mg mL^{-1}) with a determination coefficient of 0.9998.

2.6. Production of optimized HNADES-annatto seed extract

Based on the extraction conditions selected from the optimization study, annatto seed extracts were produced using the HNADES C12:C8 1:3 system. The same extraction conditions were subsequently applied to ethanol and methanol, which were used as control solvents for comparison of extraction efficiency, bioactive compound content, stability, and antimicrobial activity.

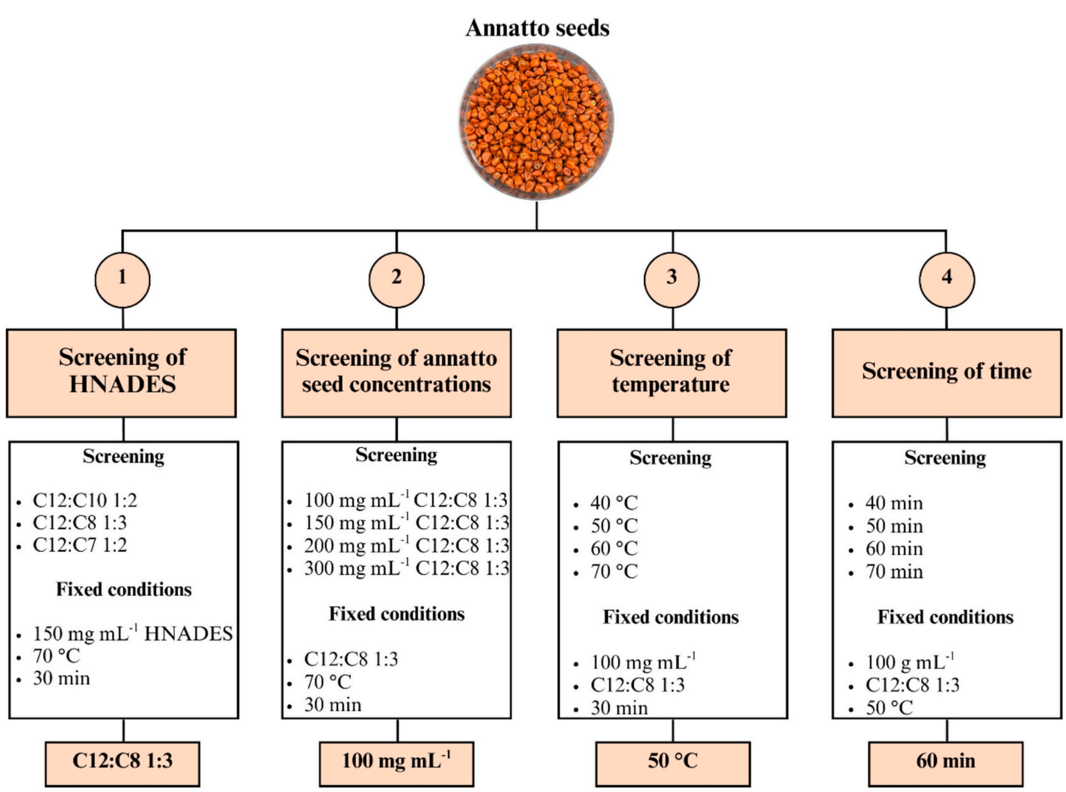

Fig. 1. Schematic diagram illustrating the optimization of extraction conditions for bioactive compounds from annatto seeds using hydrophobic natural deep eutectic solvents (HNADES), based on a one-factor-at-a-time approach.

2.7. Characterization of optimized HNADES-annatto seed extract

2.7.1. Quantification of bixin and geranylgeraniol by high performance liquid chromatography (HPLC)

Bixin and geranylgeraniol contents were determined by high-performance liquid chromatography (HPLC) using a Shimadzu system equipped with an SCL-10AVP system controller, LC-6 CE solvent delivery pumps, and an SPD-M10AVP photodiode array detector (Shimadzu Corporation, Kyoto, Japan). Annato seed extracts were diluted 20-fold in methanol and filtered through a regenerated cellulose (RC) syringe filter (13 mm, 0.22 μm). Quantification was performed using a Hypersil ODS column (250 × 4.6 mm, 5 μm) (Agilent Technologies, Santa Clara, CA, USA) with an injection volume of 20 μL, at 25 °C.

For bixin quantification, the mobile phase consisted of water/formic acid (98/2, *v*/v; solvent A) and methanol/formic acid (98/2 v/v; solvent B), at a flow rate of 0.9 mL min^{-1}, following the method described by [6]. The gradient program was as follows: 30% B in initial conditions; 60% B (15 min); 80% B (25 min); 95% B (35 min); 95% B (45 min); Stop (60 min). A bixin calibration curve was constructed in the range of 3.2–50 μg mL^{-1} ($R^2 = 0.9999$). Detection was performed at 459 nm, and results were expressed as μg bixin g^{-1} of annatto seed.

For geranylgeraniol quantification, the mobile phase consisted of water/formic acid (99.9/0.1, v/v; solvent A) and methanol (solvent B), at a flow rate of 1.0 mL min^{-1}. The gradient program was as follows: 90% B at initial conditions; 100% B (15 min); 100% B (20 min); 90% B (25 min), Stop (35 min). A geranylgeraniol calibration curve was constructed in the range of 15.6–250 μg mL^{-1} ($R^2 = 0.9936$). Detection was performed at 207 nm, and results were expressed as μg geranylgeraniol g^{-1} of annatto seed.

2.7.2. Stability of total carotenoid content, antioxidant capacity, and color

The stability of total carotenoid content, antioxidant capacity, and color parameters of optimized HNADES–annatto seed extract and controls were monitored over 90 days (0, 3, 9, 16, 30, 60, and 90 days) at 4 °C under dark conditions. The total carotenoid content and antioxidant capacity were evaluated according to the methodologies described in Sections 2.5.1. and 2.5.2, respectively. Color parameters L^* (lightness), a^* (red/green), and b^* (blue/yellow) in the CIELab color space were determined from digital images acquired using an iPhone 11 camera (Apple Inc., Cupertino, CA, USA) and processed with ImageJ software (version 1.54 g, bundled with 64-bit Java 1.8.0, January 2026). Digital images were acquired using an iPhone 11 camera under standardized conditions, including controlled LED illumination, a fixed camera-to-sample distance, and a uniform white background. Image capture was performed inside a light-controlled box to minimize shadows and external light variability, ensuring consistent color measurements. The hue angle (h^*), which represents the chromatic tone of the samples, was calculated according to Eq. (2).

$$h^* = arctan\left(\frac{b^*}{a^*}\right) \quad (2)$$

2.7.3. Antimicrobial activity against foodborne bacteria: diffusion assay, minimal inhibitory concentration (MIC), and minimum bactericidal concentration (MBC)

The antimicrobial activity of annatto seed-based extracts was evaluated using a disc diffusion assay against foodborne bacteria *Escherichia coli*, *Salmonella* sp. ATCC 14029, *Pseudomonas aeruginosa* ATCC 27853, *Campylobacter jejuni* ATCC 33560, and *Staphylococcus aureus*. For application of the annatto seed–based extracts to inoculated agar, sterile paper discs were immersed in the extracts. The sensitivity of the discs against the tested bacteria was classified according to the inhibition zone diameters (mm) as follows: not sensitive for diameters <8 mm, sensitive for diameters from 9 to 14 mm, very sensitive for diameters from 15 to 19 mm, and extremely sensitive for diameters >20 mm, adapted from Hajji et al. [34]. A sterile paper disc containing sterile water was used as a negative control, whereas discs of commercial antibiotic – Azithromycin (15 μg per disc), Ciprofloxanine (5 μg per disc) and Imipenem (10 μg per disc) – were used as a positive control.

The determination of the MIC and MBC of the annatto seed extracts against the same bacterial strains were performed using the Müller-Hinton broth microdilution method as described by Hosseini et al. [35] with adaptations. An aliquot of 100 μL of overnight bacterial suspensions at 1×10^5 CFU mL^{-1} (standardized with a McFarland scale 0.5) was added to each well. Serial dilutions of annatto-based extracts and solvents as controls (C12:C8 1:3, ethanol, and methanol) were prepared in 96-well plates, with 100 μL in the first well along with 100 μL of inoculum. After incubation at 37 °C for 24 h, 20 μL of 3% tetrazolium solution was added to each well. The first well without bacterial growth (no red color formation) was reported as the MIC (mg mL^{-1}). For MBC determination, a sterile swab was immersed, beginning with the two concentration wells preceding the first well that exhibited no bacterial growth and continuing through the wells devoid of visible growth, and then transferred to sterile Müller-Hinton agar plates, which were incubated for 24 h at 37 °C. The lowest concentration corresponding to plates with no colony growth was reported as the MBC (mg mL^{-1}).

2.7.4. Greenness assessment

The greenness of the proposed HNADES-annatto seed extraction method was evaluated using the AGREEprep metric [36], which assesses the environmental sustainability of sample preparation based on the ten principles of Green Analytical Chemistry. The evaluation was performed using the AGREEprep software by selecting the parameters corresponding to the proposed extraction procedure. The final score ranges from 0 to 1, with higher values indicating a greener sample preparation method.

2.8. Statistical analysis

All experiments were performed in triplicate, and results are presented as mean ± standard deviation. Statistical analysis of the data was carried out using Statistica software (version 7, StatSoft Inc., Hamburg, Germany). Differences among means were assessed by one-way analysis of variance (ANOVA), and Tukey's multiple comparison test was applied to identify significant differences at a significance level of $\alpha = 0.05$.

3. Results and discussion

3.1. Characterization of HNADES

3.1.1. Density and apparent acidity

Physicochemical properties of solvents are important parameters that can influence the extraction of bioactive compounds. The solvent density can affect diffusion, solvent-solute interactions, and solvent penetration at the solid-solute interface [17]. The HNADES presented similar density values ($p > 0.05$) around 0.905 g cm^{-3} (Table 1). These densities were higher than conventional solvents studied and consistent with previously reported values for C12:10 1:2, C12:C9 1:3 and C12:C8 1:3, which ranged from 0.900 to 0.905 g cm^{-3} at 25 °C [17]. HNADES formulated with C12 and other fatty acids, such as 1-octanol, 1-hexanol and decyl alcohol, also exhibit densities around 0.910 g.cm^{-3} [37]. Moreover, the densities of HNADES were lower than those of hydrophilic NADES, which typically presented values around 1.15 g cm^{-3} at 25 °C [22]. This behavior can be explained to the longer carbon chains of the HNADES constituents, resulting in a less compact molecular structure and greater free volume [37].

The apparent acidity of the HNADES is another physicochemical characteristic that may be relevant for their application as extraction solvents. The estimated values showed no major differences among the HNADES, indicating a similarly acidic character, whereas the conventional organic solvents exhibited values close to neutrality (Table 1). Because the conventional pH scale is strictly defined for aqueous

Table 1
Physicochemical properties of hydrophobic natural deep eutectic and conventional solvents at 25 °C.*

Solvent	Molar ratio	Density ($g\ cm^{-3}$)	pH	Apparent viscosity at 1 s^{-1} (mPa s)	Apparent viscosity at 1 s^{-1} and 50 °C (mPa s)
C12:C10	1:2	0.899 ± 0.009^{a}	~ 5.0	12.87 ± 0.45^{a}	5.68 ± 0.09^{a}
C12:C8	1:3	0.903 ± 0.007^{a}	~ 4.5	6.96 ± 0.12^{b}	3.97 ± 0.03^{b}
C12:C7	1:2	0.914 ± 0.002^{a}	~ 5.0	6.16 ± 0.25^{c}	3.60 ± 0.08^{c}
Ethanol	100%	0.800 ± 0.001^{b}	6.74 ± 0.01^{a}	–	–
Methanol	100%	0.805 ± 0.003^{b}	6.52 ± 0.02^{b}	–	–

* Mean ± standard deviation. Different letters (a-c) in the same column indicate significant differences between the means of the different samples, according to the Tukey test ($p < 0.05$). C12, C10, C8, and C7: dodecanoic, decanoic, octanoic, and heptanoic acids, respectively.

systems, the values obtained using pH indicator strips should be interpreted only as qualitative indicators of the apparent acidity of the HNADES rather than as absolute thermodynamic pH values [38]. Nevertheless, pH indicator strips have been reported as a simple qualitative tool for estimating the apparent acidity of DES-like systems, allowing comparative evaluation among formulations prepared and measured under the same conditions. The apparent acidity values obtained in this study were slightly higher than those reported for C12:C10 1:2, C12:C9 1:3, and C12:C8 1:3 (around 2.8) [18], as well as those reported for some common hydrophilic NADES, such as choline chloride combined with lactic acid, tartaric acid, or glycerol [22]. Although these values should not be directly compared with conventional aqueous pH values, they indicate that none of the evaluated HNADES presented an extremely acidic environment.

3.1.2. Viscosity

Viscosity is another crucial physicochemical parameter for the application of HNADES in the extraction yield of bioactive compounds, as it can often act as a limiting factor. Viscosity can be affected by the hydrogen-bond donor and acceptor species, the molar ratio of the eutectic mixture, and, in the case of hydrophilic NADES, by the amount of added water [22]. HNADES with high viscosity may reduce molecular mobility, affect mass transfer, and consequently decrease the extraction yield of bioactive compounds by limiting solid–solvent interactions [39].

The viscosity-temperature relationship of the HNADES was determined (Fig. S1a). As expected, increasing the temperature led to a decrease in apparent viscosity, a behavior also observed in other studies [14,18,40]. This occurs because increasing the temperature raises the system's energy and, consequently, the molecular motion. The higher molecular motion reduces particle–particle interactions within the solvent, thereby lowering friction and internal resistance. As a result, the viscosity of the HNADES decreases as temperature increases [18]. At 25 and 50 °C, the apparent viscosity of C12:C10 1:2 was higher than that of C12:C8 1:3 and C12:C7 1:2 ($p < 0.05$), while C12:C7 exhibited the lowest one ($p < 0.05$) (Table 1). As mentioned, the viscosity of HNADES is also influenced by their constituents and increasing the alkyl chain length of HBD reduced the viscosity of the eutectic system at the same temperature [19]. At 50 °C, these apparent viscosity values were similar to those reported for C12:C10 1:2, C12:C9 1:3, and C12:C8 1:3 [17].

3.1.3. Polarity

One of the most important characteristics of HNADES, if not the most crucial, is polarity, which directly affects the extraction yield of bioactive compounds based on the principle that "like dissolves like". Polarity can be estimated using an indirect solvatochromic method, which calculates the E_{NR} of Nile Red dye diluted in different solvents by analyzing the wavelength of maximum absorbance [29].

The HNADES exhibited lower polarity, as indicated by higher E_{NR} values, than conventional solvents ($p < 0.05$), with the C12:C8 1:3 system showing the lowest polarity among them ($p < 0.05$) (Fig. S1b). The lower polarity of the C12:C8 1:3 compared to the other HNADES may result from the higher molar fraction of the shorter-chain fatty acid (C8) in this formulation relative to the other systems. Indeed, the E_{NR} values of the HNADES were closer to that of hexane, around 54 kcal mol^{-1} [41,42], a highly nonpolar solvent, highlighting their potential for carotenoid extraction since these compounds are lipophilic. In contrast, the conventional solvents showed intermediate polarity, with methanol being the most polar among them ($p < 0.05$), although still less polar than water, which presented an E_{NR} value close to 49 kcal mol^{-1} [43]. The E_{NR} values obtained for the fatty acid-based HNADES were also consistent with those reported for other hydrophobic eutectic systems, confirming their low-polarity character [8,37,42–44]. Although ethanol and methanol were used in this study as conventional reference solvents for annatto seed extraction, their intermediate polarity contrasts with the hydrophobic nature of the HNADES. Consequently, the lower polarity of the C12:C8 1:3 system is expected to favor the solubilization of lipophilic compounds, approaching the polarity of conventional nonpolar solvents such as hexane [42].

3.1.4. Differential scanning calorimetry (DSC)

A common definition of NADES is that they are formed by mixing two or more natural components which, at a specific molar ratio, show a reduced melting temperature compared to the individual components and behave as a pure substance [19]. Therefore, the thermal properties of the fatty acids and HNADES were evaluated to verify that the prepared systems exhibited melting behavior consistent with the previously reported thermodynamic characterization of these eutectic mixtures [20].

The melting temperatures of the HNADES were lower ($p < 0.05$) than those of their individual constituents (Table 2), as can be visually observed by the endothermic peaks (Fig. 2). An increase in alkyl chain length of fatty acids led to higher melting temperatures and greater enthalpy associated with the phase transition. The same behavior was observed for HNADES: the longer the alkyl chain of the HBD, the higher the melting temperature and enthalpy (Table 2). This trend can be

Table 2
Melting temperature (Tm) and melting enthalpy (ΔH) of dodecanoic, decanoic, octanoic, and heptanoic acids (C12, C10, C8, and C7, respectively) and their corresponding hydrophobic natural deep eutectic solvents.*

Sample	Molar ratio	Tm_1 (°C)	ΔH_1 ($J\ g^{-1}$)	Tm_2 (°C)	ΔH_2 ($J\ g^{-1}$)
C12	–	45.85 ± 0.38^{a}	174.03 ± 2.80^{a}	–	–
C10	–	32.99 ± 0.57^{b}	157.49 ± 4.96^{b}	–	–
C8	–	16.63 ± 0.62^{d}	141.17 ± 2.62^{c}	–	–
C7	–	−7.62 ± 0.11^{g}	111.56 ± 0.74^{f}	–	–
C12:C10	1:2	22.49 ± 0.21^{c}	141.52 ± 0.01^{c}	–	–
C12:C8	1:3	8.95 ± 0.02^{f}	131.65 ± 1.58^{d}	–	–
C12:C7	1:2	−12.89 ± 0.26^{h}	122.05 ± 1.65^{e}	18.95 ± 0.64	45.22 ± 0.67

* Mean ± standard deviation. Different letters (a-h) in the same column indicate significant differences between the means of the different samples, according to the Tukey test ($p < 0.05$).

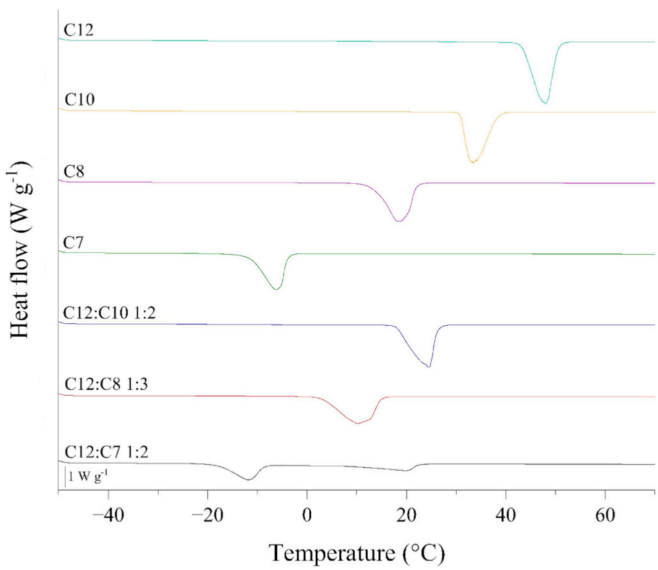

Fig. 2. Differential Scanning Calorimetry curves (Exo up) of dodecanoic, decanoic, octanoic, and heptanoic acids (C12, C10, C8, and C7, respectively) and their corresponding hydrophobic natural deep eutectic solvents.

explained by the fact that longer alkyl chains favor stronger intermolecular interactions, particularly van der Waals forces and, when applicable, hydrogen bonding, resulting in greater stabilization of the carbon chains [19]. The increase in melting temperature and enthalpy associated with longer alkyl chains was also observed for the fatty acids C12, C10, C9, and C8, as well as for the corresponding HNADES formulated using C12 as the HBA and the other fatty acids as HBDs [18,20]. Additionally, the melting temperatures of both the fatty acids and the HNADES were consistent with previously reported values [19,45,46]. These results further support the formation of the eutectic system.

3.2. Optimization of HNADES-annatto seed extract production

3.2.1. Effect of HNADES

First, the effect of the HNADES type on total carotenoid content and antioxidant capacity of the obtained extracts was investigated (Fig. 3a). The results showed that C12:C8 1:3 was the most effective system for extracting the bioactive compounds from annatto seeds, exhibiting higher total carotenoid content and antioxidant capacity ($p < 0.05$) than the extracts produced with C12:C7 1:2 and C12:C10 1:2. The higher extraction capacity of the C12:C8 1:3 system is likely associated with the combined effect of its lower polarity, which increases its affinity for lipophilic bioactive compounds, and its intermediate apparent viscosity, which may provide a favorable balance between mass transfer and stabilization during extraction (Table 1 and Fig. S1b, Supplementary material). Consequently, compared with C12:C10 1:2, the lower viscosity of C12:C8 1:3 may have facilitated mass transfer, while its higher viscosity relative to C12:C7 1:2 may have contributed to the stabilization of the extracted bioactive compounds.

Different HNADES composed of various combinations of C12, C10, C9, and C8 have been reported to exhibit distinct extraction yields of β-carotene from pumpkin, which the authors attributed to differences in carotenoid solubility in each solvent system [16]. In that study, the C10:C8 1:3 formulation showed the highest extraction yield and was therefore considered the most suitable for optimizing extraction conditions. In a similar context, the C12:C8 1:3 system was selected in the present study as the optimal eutectic mixture to further optimize the extraction conditions.

3.2.2. Effect of solid-to-solvent ratio

Regarding the effect of the solid-to-solvent ratio, the ratio of 100 mg annatto seed mL^{-1} C12:C8 1:3 exhibited the highest total carotenoid content and antioxidant capacity ($p < 0.05$) (Fig. 3b). Further increases in the solid-to-solvent ratio resulted in a gradual decrease ($p < 0.05$) in bioactive properties. This trend is consistent with diffusion-driven extraction kinetics, in which solvent saturation limits the concentration gradient and reduces mass transfer [28].

Similar trends have been reported in studies using different plant matrices and solvent systems, in which a solid-to-solvent ratio of 100 mg mL^{-1} has been identified as optimal for the extraction of bioactive compounds. Examples include the extraction of bixin from annatto seed using NADES formed by zinc chloride and diethylene glycol [47], the extraction of carotenoids from orange peel using different HNADES [8], and the extraction of phenolic compounds from leaves using aqueous and hydroethanolic solvents [48]. Additionally, a previous study demonstrated that a solid-to-solvent ratio of 100 mg mL^{-1} was the most efficient for extracting bixin from annatto seeds using sodium hydroxide as the solvent [13]. This indicates that this solid-to-solvent ratio appears to provide the best balance between extraction efficiency and solvent saturation, ensuring efficient mass transfer while avoiding solvent saturation, increased viscosity, and diffusional limitations that can impair solubilization and reduce extraction efficiency at higher matrix concentrations [28]. Therefore, the solid-to-solvent ratio of 100 mg mL^{-1} was selected for further optimization of the extraction conditions.

3.2.3. Effect of temperature

Temperature is one of the main factors affecting the extraction yield of bioactive compounds, influencing both extraction efficiency and the degradation of the extracted compounds. The HNADES-annatto seed extract obtained at 50 °C exhibited the highest total carotenoid content and antioxidant capacity ($p < 0.05$) (Fig. 3c). These parameters gradually decreased with increasing extraction temperature, with the lowest values observed at 70 °C.

Carotenoids are intracellular compounds in plant tissues, and moderate increases in extraction temperature enhance solvent penetration and mass transfer by increasing cell wall permeability, reducing solvent viscosity, and improving diffusion, provided that thermal degradation is avoided [49]. Furthermore, increasing the system temperature raises the kinetic energy of the components, thereby accelerating the extraction rate [50]. However, carotenoids are highly susceptible to enzymatic and nonenzymatic oxidation, oxygen exposure, and thermal and photo-induced degradation. Excessive temperatures during extraction may therefore accelerate carotenoid degradation and promote trans–cis isomerization, leading to color loss and reduced biological activity [51]. This explains why extraction at 50 °C yielded higher carotenoid content and antioxidant capacity, whereas higher temperatures resulted in reduced values (Fig. 3c).

Several studies have reported optimal extraction temperatures for carotenoids, particularly bixin, from annatto seeds in the range of 40 to 70 °C using different solvent systems. For instance, the optimal temperature for bixin extraction from annatto seeds using water in a double-jacketed chamber with pressure control was reported as 40 °C [52]. When methanol was used as the solvent under thermostatic bath heating, the highest bixin extraction was achieved at 45 °C [12] and 70 °C [30]. In ultrasonic-assisted extraction employing choline chloride–lactic acid–based NADES, an extraction temperature of 50 °C has been reported [40]. Therefore, carotenoid extraction yield and stability depend not only on temperature but also on the extraction method and solvent system. In the present study, 50 °C was selected as the optimal temperature for further optimization of the extraction parameters.

3.2.4. Effect of time

In addition to temperature, extraction time is another key parameter to be optimized to improve the yield of bioactive compounds while minimizing energy consumption and process costs [49]. An extraction

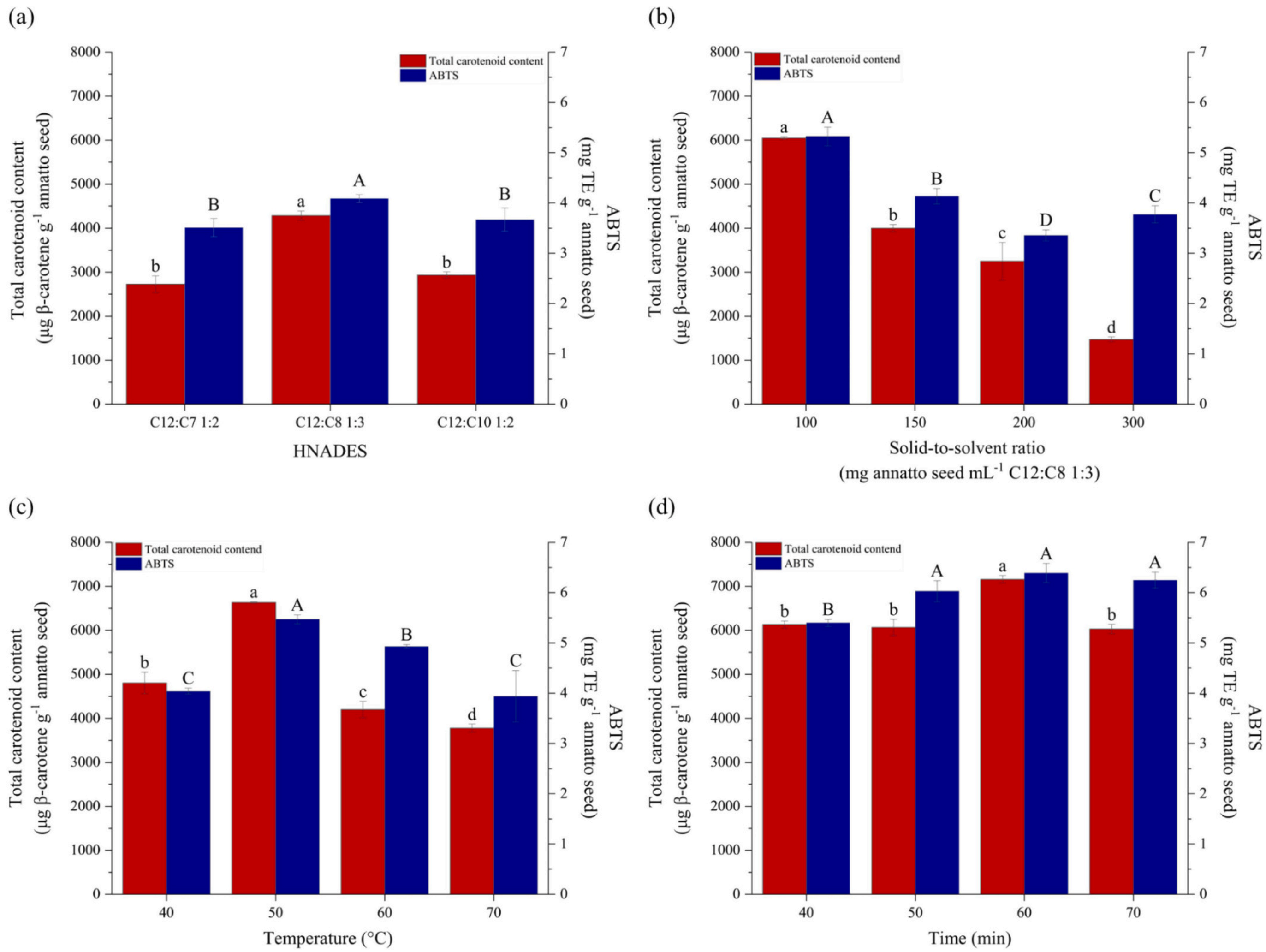

Fig. 3. Total carotenoid content and antioxidant capacity (ABTS assay) obtained during the optimization of HNADES–annatto seed extraction conditions using the one-factor-at-a-time approach: (a) HNADES type, (b) solid-to-solvent ratio, (c) extraction temperature, and (d) extraction time. Different letters indicate significant differences between the means of the different samples, according to the Tukey test ($p < 0.05$). TE: Trolox equivalent.

time of 60 min resulted in the HNADES-annatto seed extract with the highest ($p < 0.05$) total carotenoid content and antioxidant capacity simultaneously (Fig. 3d). At times shorter and longer than 60 min, total carotenoid content was significantly lower ($p < 0.05$), whereas antioxidant capacity remained similar at extraction times above 50 min ($p > 0.05$). This nonlinearity may be attributed to the co-extraction of other bioactive compounds present in annatto seeds, such as geranylgeraniol, additional terpenoids, and phenolic compounds, which likely possess distinct diffusion behaviors and solvent affinities relative to carotenoids [51].

Many studies agree that increasing extraction time enhances the yield of bioactive compounds by improving the extraction rate and diffusion coefficients [30,49,50,53]. However, this effect is limited, as diffusion eventually becomes saturated and bioactive compounds either stabilize or undergo degradation [28,50]. The literature reports optimized extraction times for carotenoids from annatto seeds ranging from 10 to 180 min [12,30,52]. These variations reflect differences in extraction methods and solvent systems, which influence the optimal conditions for recovering bixin and other bioactive compounds, including extraction time. In the present study, an extraction time of 60 min was selected as optimal for optimizing carotenoid extraction conditions from annatto seeds.

Based on the optimization study, the extraction conditions selected for subsequent analyses were C12:C8 1:3 as solvent, a solid-to-solvent ratio of 100 mg mL^{-1}, extraction temperature of 50 °C, and extraction time of 60 min.

3.3. Characterization of optimized HNADES-annatto seed extract

3.3.1. Bixin and geranylgeraniol contents by high performance liquid chromatography (HPLC-PDA)

The HNADES–annatto seed extract (Fig. 4a) obtained under optimized extraction conditions (C12:C8 1:3 as solvent, solid-to-solvent ratio of 100 mg annatto seed mL^{-1} C12:C8 1:3, extraction temperature of 50 °C, and extraction time of 60 min) was analyzed for its bixin and geranylgeraniol contents by HPLC-PDA (Fig. 4b). In addition, control extracts (Fig. 4a) produced under the same conditions but using ethanol or methanol as solvents were also analyzed for these compounds (Fig. 4b).

The HNADES–annatto seed extract showed higher extraction efficiency for both bixin and geranylgeraniol compared to the control extracts ($p < 0.05$), confirming the extraction potential of HNADES for nonpolar bioactive compounds. The ethanolic-annatto seed extract exhibited a higher bixin content ($p < 0.05$) than the methanolic-annatto seed extract, possibly due to its lower polarity and, consequently, greater affinity for carotenoids, whereas geranylgeraniol contents were similar for both annatto seed extracts ($p > 0.05$). Although ethanol and methanol were used as conventional reference solvents in the present

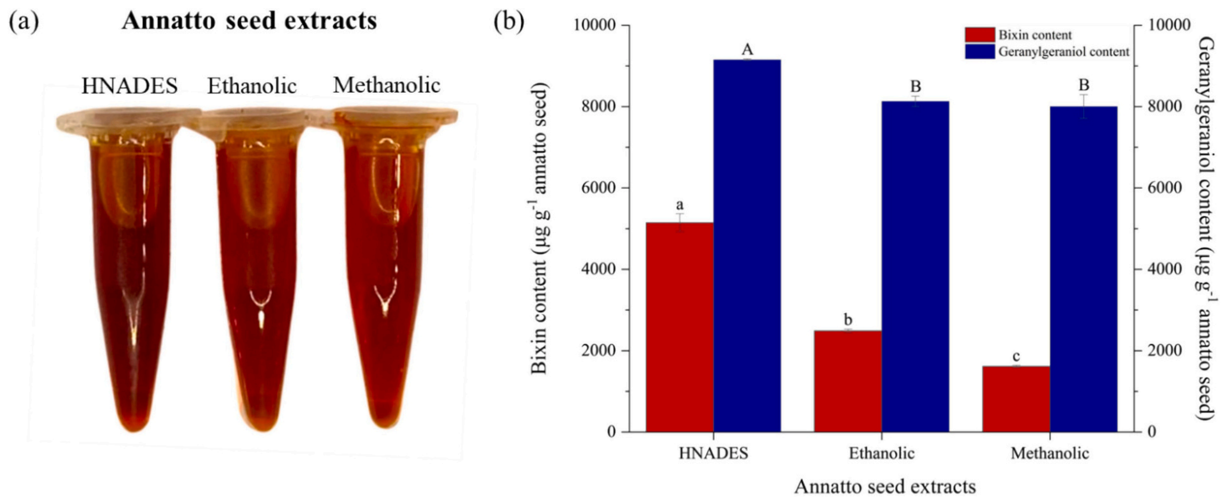

Fig. 4. (a) Photographs of the optimized HNADES, ethanolic and methanolic-annatto seed extracts, and (b) bixin and geranylgeraniol contents of these extracts. Different letters indicate significant differences between the means of the different samples, according to the Tukey test ($p < 0.05$).

study, recent studies have also demonstrated that hydrophobic deep eutectic solvents can achieve extraction yields comparable to or even higher than those obtained with conventional nonpolar solvents, such as hexane, for lipophilic bioactive compounds. These findings indicate that the superior extraction performance of HNADES cannot be attributed solely to their low polarity but also to the physicochemical characteristics of the eutectic system [42].

Interestingly, the bixin content in all annatto seed extracts was lower than that of geranylgeraniol when expressed per gram of seed. This behavior may be related to the composition and origin of the plant matrix, the developmental stage of the fruit, as well as the solubility and affinity of these compounds for the solvents, and their chemical stability [16]. Indeed, it has been reported that geranylgeraniol, an essential precursor in carotenoid biosynthesis, is the most abundant oily compound present in annatto seeds, accounting for approximately 1% of the dry seed mass [54]. Additionally, bixin is a long-chain carotenoid with an extensive conjugated double-bond system [4], which makes it highly susceptible to degradation by various agents involved in the initiation of oxidative processes [55]. In contrast, geranylgeraniol is a terpenoid with a shorter carbon chain and fewer double bonds (Martinez [10]), which may render it more resistant to degradation.

These results differed substantially from those reported in the literature for geranylgeraniol. In one study, ethanolic extracts from annatto seeds were obtained using a sonicator at different acoustic power levels and extraction times [2]. Bixin content in the extracts ranged from approximately 3500 to 9000 µg g^{-1} annatto seed, which is consistent with our results and indicates that extraction efficiency is strongly influenced by the solvent and extraction method. In contrast, the amount of geranylgeraniol extracted ranged from approximately 0.3 to 2 µg g^{-1} annatto seed [2], which is markedly lower than the values reported in the present study. Differences in chromatographic conditions, calibration strategies, and solvent selectivity may partially explain these variations.

A study specifically addressing the extraction of geranylgeraniol from annatto seeds has also been reported using nonpolar solvents, including hexane, acetone, and ethyl acetate, with concentrations of approximately 1700 µg g^{-1} [10]. This discrepancy may be attributed to the extraction efficiency of HNADES not only toward carotenoids but also toward terpenoids. In addition to the factors discussed above, the source of the plant raw material may also play a decisive role in the extraction of bioactive compounds.

To the best of our knowledge, studies involving fatty acid–based HNADES have not yet been used to extract bixin and other bioactive compounds from annatto seeds, which makes this study particularly promising. Higher amounts of bixin and geranylgeraniol were extracted using C12:C8 1:3 as the solvent, compared with a previous study employing a NADES based on choline chloride and lactic acid, which extracted approximately 21 µg of bixin g^{-1} of annatto seed [40]. Furthermore, a NADES based on zinc chloride and diethylene glycol extracted approximately 52 mg of bixin g^{-1} dry annatto seed [47], highlighting the potential of these solvents.

3.3.2. *Stability of total carotenoid content, antioxidant capacity, and color*

As previously mentioned, carotenoids, such as bixin, are susceptible to degradation when exposed to environmental variations, including temperature, light, and pH, due to their conjugated double-bond system [55,56]. Although geranylgeraniol and other bioactive compounds may appear to be less unstable, they are also highly susceptible to oxidative degradation. Therefore, it is essential to evaluate the chemical stability of these compounds under storage conditions. Visually, all annatto seed extracts exhibited a less intense red coloration after 90 days of storage (Fig. 5a). The loss of color intensity is one of the earliest indicators of carotenoid degradation and was associated with total carotenoid content [52], antioxidant capacity, and color tone measured over time.

The total carotenoid content and antioxidant capacity of all annatto seed extracts decreased over storage time (Fig. 5b,c), as expected. Despite storage under light-protected conditions and at low temperature, the presence of gaseous O_2 in the storage environment may promote the oxidation of bioactive compounds [57,58]. Moreover, structural rearrangements, and trans–cis isomerization, particularly in the case of carotenoids, may further contribute to their degradation over time [49,52,55]. Other studies have also reported a progressive reduction in total carotenoid content, antioxidant capacity, and/or bixin content in extracts over storage time [11,56,57].

Nevertheless, after 90 days of storage, the reduction observed for the HNADES–annatto seed extract was approximately 30% for both total carotenoid content and antioxidant capacity, showing improved stability compared to controls from day 30 of storage onward. In contrast, ethanolic and methanolic-annatto seed extracts exhibited markedly higher losses after 90 days, with reductions of approximately 65% in antioxidant capacity for both solvents and decreases of about 65% and 75% in total carotenoid content for ethanolic and methanolic-annatto seed extracts, respectively. The greater degradation of bioactive compounds in conventional solvents may be linked to the higher oxygen solubility in these systems, which can promote oxidative reactions [59]. In contrast, although oxygen may dissolve in HNADES, its diffusional mobility is likely constrained by their high viscosity [60], potentially slowing oxidation processes.

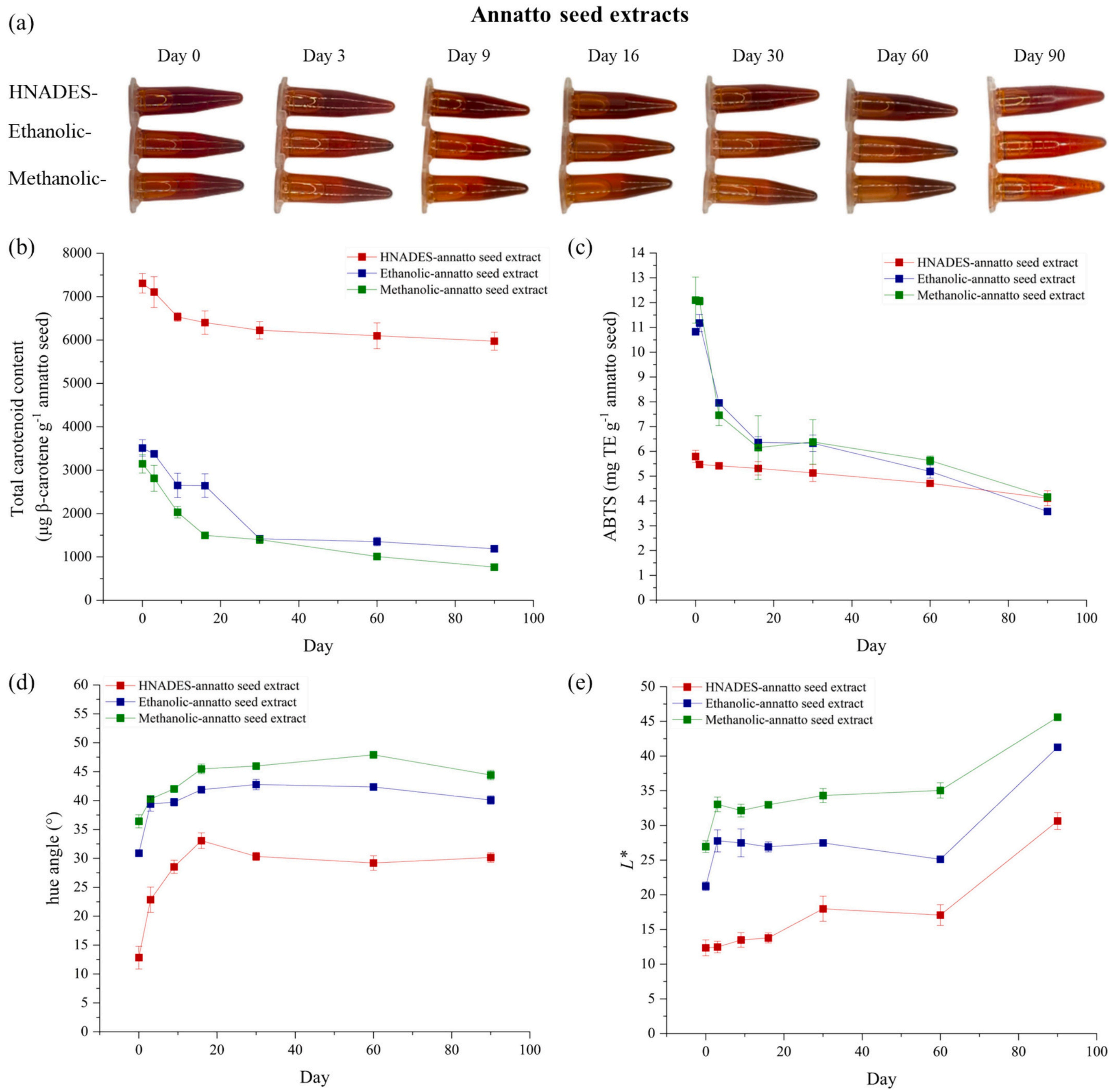

Fig. 5. (a) Photographs of HNADES, ethanolic, and methanolic-annatto seed extracts and the stability of (b) total carotenoid content, (c) antioxidant capacity, (d) hue angle, and (e) lightness (L*) over 90 days of storage. TE: Trolox equivalent.

Together, these findings position fatty acid–based HNADES not only as extraction media but also as functional stabilization systems for oxidation-sensitive bioactives. Indeed, previous studies have reported that NADES and HNADES are able to preserve the biological activities of plant extracts rich in bioactive compounds more efficiently than conventional solvents, as demonstrated for carotenoids and polyphenols extracted from orange peel using HNADES [8], and anthocyanins extracted from bilberry using NADES [61]. This behavior can be attributed to the intrinsic properties of hydrophobic eutectic systems, such as thermal stability, high viscosity, and chemical interactions among fatty acid chains [8]. As a result, bioactive compounds may become physically confined within the HNADES network, while their mobility is further limited by the high viscosity of the system [15], thereby reducing their exposure to gaseous O_2 [61]. Therefore, the stabilizing effect of the HNADES is likely associated with the physicochemical characteristics of the eutectic matrix rather than with a direct preservative or antioxidant action of the solvent itself.

Another important characteristic to be evaluated during the storage of pigment-rich extracts is color. The hue angle, calculated from the a^* and b^* color parameters, represents the specific color range of the extract [30], and changes in this parameter can be correlated with the degradation of colored compounds, such as bixin, as well as with differences in bixin isomeric structures and norbixin content. Bixin-rich extracts exhibit red, orange, and yellow hues, with hue angle values typically ranging between 20 and 50° [30,40,52], where lower values closer to zero indicate a more intense red coloration.

Initially, the HNADES–annatto seed extract exhibited a hue angle of approximately 13°, indicating an intense red coloration, whereas the ethanolic and methanolic-annatto seed extracts showed hue angle values around 35°, corresponding to a red-orange tone that was still intense but less pronounced than that of the HNADES-annatto seed extract (Fig. 5d). This observation corroborates the bixin content results (Fig. 4b), in which the HNADES–annatto seed extract exhibited significantly higher values than the other extracts. This difference may also be explained by a higher relative content of more stable isomer trans-bixin (red colored) in the HNADES–annatto seed extract [2], whereas cis-bixin exhibits a redder–orange coloration, in addition to possible differences in norbixin content among the extracts [52].

Over storage time, the hue angle of all annatto seed extracts increased, as expected due to the degradation of bixin and other carotenoids (Fig. 5b), leading to a reduction in the red color intensity. From day 30 of storage onward, the hue angle tended to stabilize at approximately 30° for the HNADES–annatto seed extract, 41° for the ethanolic-annatto seed extract, and 45° for the methanolic-annatto seed extract. This behavior suggests that the loss of antioxidant capacity (Fig. 5c) after 30 days of storage may be more strongly associated with the degradation of other bioactive compounds, such as geranylgeraniol and related compounds, rather than with bixin and other minor carotenoids. Future studies monitoring the stability of individual terpenoids during storage could further clarify this behavior.

On day 90 of storage, although the hue angle values remained practically stable, all extracts exhibited a pronounced increase in the lightness parameter (L^*) (Fig. 5e), which is visually evident in Fig. 5a, indicating that the samples became lighter and likely contained a lower concentration of pigments. In comparison with other studies, the hue angle values obtained at day 90 of storage were comparable to those reported for annatto seed extracts prepared using conventional solvents, such as water (44°) [52], choline chloride:lactic acid-based NADES (51°) [40], ethanol (39°) [2], ethanol 60% (26°) [40], and methanol (25°) [12].

Overall, these results highlight the potential of HNADES as an extraction and stabilization medium for annatto seed pigments, as it provided higher initial color intensity, greater retention of the red hue possibly associated with trans-bixin, and improved color stability during storage compared to conventional solvents. Although the present study demonstrated the stabilizing effect of HNADES under refrigerated and light-protected storage conditions, future studies should evaluate the stability of these systems under different environmental conditions, including exposure to light, oxygen, and elevated temperatures [62], to further assess their practical applicability.

3.3.3. Antimicrobial activity against foodborne bacteria: diffusion assay, MIC, and MBC

In the antimicrobial susceptibility diffusion assay, most of the tested bacteria showed sensitivity to the ethanolic-annatto seed extract, followed by the methanolic-annatto seed extract and the HNADES-annatto seed extract (Table S1). Among the solvent controls, methanol did not affect the growth of *Salmonella* sp. and *E. coli*, whereas ethanol did not affect the growth of any of the tested bacteria, and C12:C8 1:3 affected only the growth of *C. jejuni*. These results indicate that, depending on the solvent used for extraction, different antimicrobial compounds may be recovered and exhibit activity against distinct microorganisms. Furthermore, they suggest that the solvent itself may exert a synergistic effect on antimicrobial activity.

The HNADES-annatto seed extract exhibited a pronounced effect against *C. jejuni*, which may be partly associated with the intrinsic activity of the solvent system. The antimicrobial activity of HNADES based on different fatty acids has already been reported against several Gram-positive and Gram-negative bacteria [19]. In addition, HNADES systems may effectively disrupt and remove established microbial biofilms, likely through a dual mechanism involving biofilm matrix destabilization and direct bactericidal effects.

Moreover, the HNADES-annatto seed extract showed an intermediate effect against *P. aeruginosa*, which was not sensitive to the solvent alone. This finding indicates that the antimicrobial activity against this bacterium is likely associated with the bioactive compounds present in the HNADES-annatto seed extract, such as bixin and geranylgeraniol. One study reported a lack of sensitivity of annatto seed extracts obtained using different organic solvents against *E. coli*, while inhibition was observed for *S. aureus* [63]. In contrast, another study reported sensitivity of both *S. aureus* and *Bacillus cereus* to hydroethanolic annatto seed extract [64]. It should be noted that agar diffusion assays may underestimate the antimicrobial potential of highly viscous or hydrophobic extracts, such as HNADES-based systems, due to reduced diffusion through the agar matrix. Therefore, the absence of inhibition zones does not necessarily indicate a lack of antimicrobial activity. The MIC and MBC results confirm this hypothesis (Table 3).

Despite the wide variation observed in MIC and MBC among the annatto seed extracts, it can be noted in most extracts, the MBC values were markedly higher than the MIC values, indicating a predominantly bacteriostatic effect (MBC/MIC >4). In a few cases, MBC values were lower than MIC values for solvent controls. This apparent inversion may be attributed to methodological limitations of the colorimetric assay, including possible solvent interference with tetrazolium reduction, as well as differences between metabolic inhibition and actual cell viability [65].

Notably, the HNADES-annatto seed extract and the C12:C8 1:3 system exhibited relevant antibacterial activity, as evidenced by their low MIC values against the tested microorganisms. In contrast, the higher MBC values for HNADES-annatto seed extract and the C12:C8 1:3 (17.14 and 30.21 mg mL^{-1}, respectively) suggest that bacterial cells were not eliminated at concentrations close to the MIC. Thus, both systems demonstrated good inhibitory capacity, particularly against *S. aureus*, *P. aeruginosa*, and *C. jejuni*, and may be considered to possess relevant antibacterial properties, given that the MIC values were approximately 0.5 mg mL^{-1} [66]. The MIC values obtained for the HNADES–annatto seed extract were consistent with those reported for pure bixin against different bacterial strains [35].

These results may be associated with the high contents of bixin and geranylgeraniol in the HNADES system, as these compounds have been previously reported as effective antimicrobial [9,35,63,64]. Moreover, the C12:C8 1:3 system may have enhanced the antimicrobial effect by acting synergistically with the bioactive compounds extracted from annatto seeds [19,35].

Although the ethanolic and methanolic-annatto seed extracts, as well as methanol alone, also showed a more pronounced bacteriostatic than bactericidal effect, their antimicrobial efficiency was substantially lower when compared with the HNADES-annatto seed extract and the HNADES system itself, as indicated by their higher MIC values. Overall, these findings highlight the potential use of these extracts, particularly the HNADES-annatto seed extract, as natural antimicrobial agents with inhibitory activity capable of delaying the growth of both Gram-negative and Gram-positive microorganisms.

3.3.4. Greenness assessment

The greenness of the proposed extraction method was assessed using the AGREEprep metric, yielding an overall score of 0.66 (Fig. S2, Supplementary Material). This score indicates a satisfactory level of environmental sustainability for the sample preparation procedure [36]. The favorable result is mainly attributed to the use of renewable fatty acid-based HNADES as extraction solvents and the absence of hazardous reagents during sample preparation. Overall, the AGREEprep assessment supports the proposed HNADES-based extraction as an environmentally sustainable sample preparation approach.

4. Conclusion

This study demonstrated that fatty acid–based hydrophobic natural deep eutectic solvents (HNADES) are effective multifunctional green solvents for the extraction of bioactive compounds from annatto seeds.

Table 3
Minimum inhibitory concentration (MIC) and minimum bactericidal concentration (MBC) of HNADES-, ethanolic-, and methanolic- annatto-seed extracts (ASE).

Samples	Activity (mg mL^{-1})	*Salmonella* sp.	*E. coli*	*C. jejuni*	*S. aureus*	*P. aeruginosa*
HNADES-ASE	MIC	2.07	1.97	0.52	0.49	0.50
	MBC	17.14	17.14	17.14	17.14	17.14
Ethanolic-ASE	MIC	55.14	27.57	13.78	27.60	13.78
	MBC	220.56	220.56	55.14	110.28	27.57
Methanolic-ASE	MIC	25.38	25.38	22.81	23.85	16.60
	MBC	203.06	406.11	101.53	203.06	50.76
C12:C8 1:3	MIC	3.77	1.84	0.46	3.35	0.48
	MBC	30.21	30.21	30.21	30.21	30.21
Ethanol	MIC	53.33	25.42	25.21	16.51	33.02
	MBC	26.67	26.67	26.67	26.67	26.67
Methanol	MIC	52.29	52.29	25.63	52.09	26.15
	MBC	26.15	26.15	26.15	26.15	26.15

Among the evaluated systems, the C12:C8 (1:3) formulation provided the highest carotenoid recovery, particularly for bixin, while also enhancing the stability of oxidation-sensitive compounds during storage. The improved preservation of bioactive molecules was likely associated with the physicochemical properties of HNADES, including their high viscosity and reduced oxygen diffusional mobility, which may limit oxidative degradation. Additionally, the optimized extract exhibited relevant antibacterial activity, reinforcing its potential applicability in food, pharmaceutical, and cosmetic formulations. Together, these findings position fatty acid–based HNADES not only as extraction media but also as functional stabilization systems capable of simultaneously enhancing extraction efficiency and improving the stability of lipophilic bioactives. The AGREEprep assessment further confirmed the environmental sustainability of the proposed extraction method, supporting its alignment with the principles of Green Analytical Chemistry. Therefore, this approach expands the technological potential of hydrophobic eutectic systems and supports their use in the development of high-value natural products within sustainable processing frameworks. This strategy contributes to the replacement of volatile petrochemical solvents and supports safer and more sustainable industrial extraction processes.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI) to assist with English language editing and improve the clarity and readability of the manuscript. The authors carefully reviewed and edited all AI-generated content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Larissa Tessaro: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft. **Maria Eduarda de Almeida Astolfo:** Data curation, Formal analysis, Investigation. **Julia Rabelo Vaz Matheus:** Formal analysis, Investigation, Methodology, Writing – original draft. **Severino Matias de Alencar:** Methodology, Resources, Validation, Writing – review & editing. **Ana Elizabeth Cavalcante Fai:** Methodology, Resources, Validation, Writing – review & editing. **Bianca Chieregato Maniglia:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors acknowledge the Sao Paulo Research Foundation (FAPESP) for the grants (2020/08727-0; 2024/02072-3; 2023/17299-0) and Postdoctoral fellowship of L.T. (2024/10994-8); the Brazilian National Council for Scientific and Technological Development (CNPq) for the Research fellowship of B.C.M (303423/2025-0) and Postdoctoral fellowship of L.T. (153726/2024-5; 157788/2025-3); The State Funding Agency of Rio de Janeiro (FAPERJ) for the Postdoctoral fellowship of J. R.V.M and for the Research fellowship of A.E.C.F. (E-26/200.271/2026, E-26/200.367/2025, E-26/201.428/2022, and E-26/210.326/2022). This study was financed in part by the "Coordenação de Aperfeiçoamento de Pessoal de Nível Superior–Brasil" (CAPES)-Finance Code 001.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.seppur.2026.139525.

Data availability

Data will be made available on request.

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